

■ Medical Student Data Analysis and Visualization

■ Objective

The objective of this project is to clean, analyze, and visualize data related to medical students. The project aims to identify key insights, handle missing values, and perform exploratory data analysis (EDA) to support better decision-making regarding students' academic performance, demographics, and other attributes.

■ Tools and Libraries Used

- Python
- Pandas → For data cleaning and manipulation
- NumPy → For numerical operations
- Matplotlib & Seaborn → For visualization and graphical analysis

■ Dataset Description

- Dataset Name: Medical Student Data
- File Format: CSV (Medical Student Data.csv)
- Contents: Information related to students such as demographics, academic scores, and attendance.

■ Data Cleaning Steps

1. Loaded dataset using pandas (`pd.read_csv()`).
2. Checked dataset structure using `df.head()`, `df.shape`, and `df.info()`.
3. Identified missing values using `df.isnull().sum()` and visualized with heatmap.
4. Filled missing values using `df.fillna(method='ffill')` and verified data integrity.

■ Data Visualization

- Heatmap: Visualize missing values.
- Histograms: Analyze data distribution.
- Boxplots: Identify outliers in students' marks.
- Correlation Heatmap: Understand relationships between numeric variables.
- Countplots: Analyze gender, course, and performance distributions.

■ Insights & Findings

- Identified impact of missing values.
- Found correlations between attendance and performance.
- Observed trends in academic scores across demographics.
- Detected outliers and anomalies.
- Summarized cleaned data for reporting.

■ Project Outcome

This project demonstrates practical skills in Data Cleaning and Exploratory Data Analysis (EDA). It showcases the ability to visualize and interpret real-world data, handle missing values, and derive meaningful insights using Python libraries.

■ Conclusion

The Medical Student Data project successfully transforms raw student data into clean, structured, and visual insights. The final analysis provides a foundation for data-driven decision-making in educational or healthcare contexts.